Practice 5: Conditional Probability in AI medical model

Nombre del alumno

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Objective

Simulate an artificial intelligence-based medical diagnostic system using distributions and concepts of conditional probability. Analyze how the disease prevalence, sensitivity (test's ability to detect the sick), and specificity (ability to correctly identify healthy individuals) affect the performance of the automatic diagnosis.

Configuration

Before running the simulation, key parameters must be defined that will directly influence the results of the AI-simulated diagnosis. These parameters are:

Semilla aleatoria (set.seed):

Ensures all get the same results when executing the code. It's useful for a controlled and reproducible simulation environment.

Cantidad de pacientes (n_pacientes):

Total number of individuals included in the simulation. The greater the number, the more robust the statistical results.

prevalence:

Actual probability that a person is sick in the simulated population. For example, if it is 0.01, only 1% will actually be sick.

sensitivity:

Probability that the AI correctly detects a sick person. Interpreted as the hit rate among those who are truly sick.

specificity:

Probability that the AI correctly identifies a healthy person. Interpreted as the hit rate among those who are not sick.

These initial values simulate a realistic automatic diagnostic environment and will be the foundation for calculating conditional probabilities later on:

Code:

```
set.seed(123)

n_pacientes <- 10000

prevalence <- 0.01
sensitivity <- 0.90
specificity <- 0.95
```

Simulation and AI Diagnosis

This simulates two things: Who is actually sick in our simulated population What diagnosis the AI provides for each patient

Step 1: Simulate the reality (who is sick)

Use a binomial distribution to generate a vector of patients, where each is assigned a value: 1 if they are sick, 0 if they are healthy. The probability of being sick is defined by the prevalence, which is the actual percentage of disease in the population.

Step 2: Simulate the AI's diagnosis

The AI diagnosis is also simulated with a binomial distribution: For sick patients, use the sensitivity, which is the probability that the AI correctly detects the disease. For healthy patients, use the complement of specificity (1 - specificity), which represents the probability of a false positive (AI detects disease when there is none).

Paso 3: Combinar los resultados

Create a vector to store the AI diagnoses and then assign each patient their corresponding result: Sick patients receive a simulated result based on sensitivity. Healthy patients receive a result based on the false positive probability. Code:

```
# Step 1: Simulate who is actually sick
# Use rbinom to simulate 0 (healthy) or 1 (sick) variables with
probability = prevalence
disease <- rbinom(n_pacientes, size = 1, prob = prevalence)

# Step 2: Simulate the AI's diagnosis

# a) For sick patients, AI gives a positive diagnosis with prob =
sensitivity
ai_sick <- rbinom(sum(disease == 1), size = 1, prob = sensitivity)
cat("The IA model diagnosis result is: ", ai_sick)

## The IA model diagnosis result is:  1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1
```

```
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 0 1 1 1 1 1 1 1 1 1 1 1
1 1 1 1 0 1 1 1 1

# b) For healthy patients, AI can give a false positive with prob = (1 -
specificity)
# That is, incorrectly diagnosing a healthy person as sick
ai_healthy <- rbinom(sum(disease == 0), size = 1, prob = 1 - specificity)

# Step 3: Combine the diagnoses into a full vector, aligned with all
patients
ai_diagnosis <- rep(NA, n_pacientes)  # Create an empty vector

# Assign AI results for each group
ai_diagnosis[disease == 1] <- ai_sick
ai_diagnosis[disease == 0] <- ai_healthy
```

Contingency table (reality vs AI)

```
# Cross table to compare AI diagnosis with reality
table_result <- table(Real = disease, Diagnosis = ai_diagnosis)
cat("The contigency table (reality vs AI) is: \n")

## The contigency table (reality vs AI) is:

table_result

##      Diagnosis
## Real    0    1
##    0 9397  502
##    1    7   94
```

Conditional probability calculation

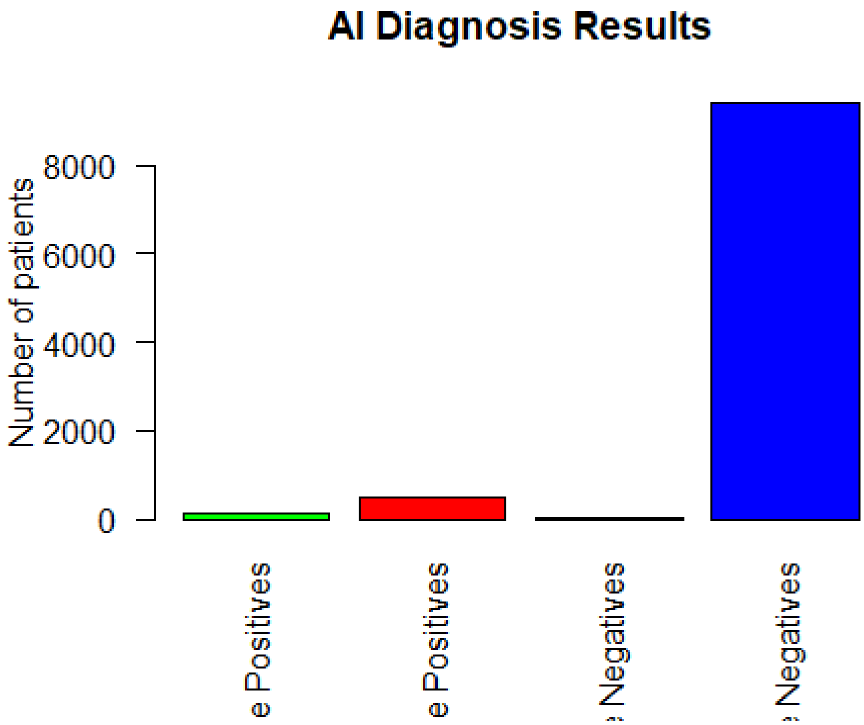
```
# Calculate the probability that a patient is actually sick
# given that the AI diagnosed them as positive: P(sick | positive
diagnosis)

# Total number of patients diagnosed as positive by the AI
total_positives <- sum(ai_diagnosis == 1)

# Of those positives, how many are truly sick (true positives)
true_positives <- sum(ai_diagnosis == 1 & disease == 1)

# Use the conditional probability formula:
# P(A|B) = P(A ∩ B) / P(B)
prob_sick_given_positive <- true_positives / total_positives
cat("The probability that a patient is actually sick is: ",
prob_sick_given_positive*100, "%")

## The probability that a patient is actually sick is:  15.77181 %
```

Análisis de errores:

```
# Number of false positives: AI says positive but patient is healthy
false_positives <- sum(ai_diagnosis == 1 & disease == 0)

# Number of false negatives: AI says negative but patient is sick
false_negatives <- sum(ai_diagnosis == 0 & disease == 1)

# Number of true negatives: AI says negative and patient is healthy
true_negatives <- sum(ai_diagnosis == 0 & disease == 0)

# Show the results
cat("The total of False positives are:", false_positives, "\n")

## The total of False positives are: 502

cat("The total of False negatives are:", false_negatives, "\n")

## The total of False negatives are: 7

cat("The total of True negatives are:", true_negatives, "\n")

## The total of True negatives are: 9397
```

Graphs:

```
# Create a bar chart with the 4 types of results

results <- c(
  "True Positives" = true_positives,
  "False Positives" = false_positives,
  "False Negatives" = false_negatives,
  "True Negatives" = true_negatives
)

# Bar chart to show the AI's hits and errors
barplot(results,
        col = c("green", "red", "orange", "blue"),
        main = "AI Diagnosis Results",
        ylab = "Number of patients",
        las = 2)
```



Activities:

1.Concatenate the printed percentage probability and add a legend when printing.
2. Change the values of prevalence, sensitivity, and specificity one time and explain how the results change. ## First change

```r
set.seed(123)

n_pacientes2 <- 100000

prevalence2 <- 0.10
sensitivity2 <- 0.90
specificity2 <- 0.95
```

2nd change

```r
# Step 1: Simulate who is actually sick
# Use rbinom to simulate 0 (healthy) or 1 (sick) variables with
probability = prevalence
disease2 <- rbinom(n_pacientes2, size = 1, prob = prevalence2)

# Step 2: Simulate the AI's diagnosis

# a) For sick patients, AI gives a positive diagnosis with prob =
sensitivity
ai_sick2 <- rbinom(sum(disease2 == 1), size = 1, prob = sensitivity2)
cat("The IA model diagnosis result is: ", ai_sick)
```

```
## The IA model diagnosis result is:  1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 0 1 1 1 1 1 1 1 1 1 1
1 1 1 1 0 1 1 1 1

# b) For healthy patients, AI can give a false positive with prob = (1 -
specificity)
# That is, incorrectly diagnosing a healthy person as sick
ai_healthy2 <- rbinom(sum(disease2 == 0), size = 1, prob = 1 -
specificity2)

# Step 3: Combine the diagnoses into a full vector, aligned with all
patients
ai_diagnosis2 <- rep(NA, n_pacientes2)  # Create an empty vector

# Assign AI results for each group
ai_diagnosis2[disease2 == 1] <- ai_sick

## Warning in ai_diagnosis2[disease2 == 1] <- ai_sick: número de
elementos para
## sustituir no es un múltiplo de la longitud del reemplazo

ai_diagnosis2[disease2 == 0] <- ai_healthy

## Warning in ai_diagnosis2[disease2 == 0] <- ai_healthy: número de
elementos para
## sustituir no es un múltiplo de la longitud del reemplazo
```

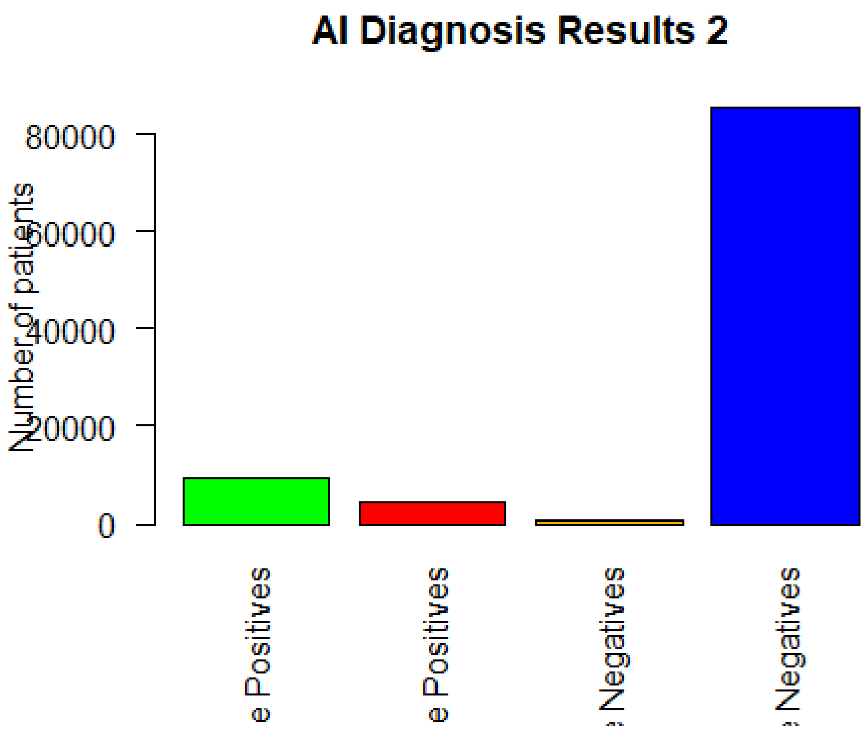
##Table

```
# Cross table to compare AI diagnosis with reality
table_result2 <- table(Real = disease2, Diagnosis = ai_diagnosis2)
cat("The contigency table (reality vs AI) is: \n")

## The contigency table (reality vs AI) is:

table_result2

##     Diagnosis
## Real     0     1
##    0 85493  4568
##    1   688  9251
```

3. Interpret the conditional probability obtained: is the positive diagnosis reliable?

```
# Calculate the probability that a patient is actually sick
# given that the AI diagnosed them as positive: P(sick | positive
diagnosis)

# Total number of patients diagnosed as positive by the AI
total_positives2 <- sum(ai_diagnosis2 == 1)
```

```r
# Of those positives, how many are truly sick (true positives)
true_positives2 <- sum(ai_diagnosis2 == 1 & disease2 == 1)

# Use the conditional probability formula:
# P(A|B) = P(A ∩ B) / P(B)
prob_sick_given_positive2 <- true_positives2 / total_positives2
cat("The probability that a patient is actually sick is: ",
prob_sick_given_positive2*100, "%\n")

## The probability that a patient is actually sick is:  66.94406 %

# Number of false positives: AI says positive but patient is healthy
false_positives2 <- sum(ai_diagnosis2 == 1 & disease2 == 0)

# Number of false negatives: AI says negative but patient is sick
false_negatives2 <- sum(ai_diagnosis2 == 0 & disease2 == 1)

# Number of true negatives: AI says negative and patient is healthy
true_negatives2 <- sum(ai_diagnosis2 == 0 & disease2 == 0)

# Show the results
cat("The total of False positives are:", false_positives2, "\n")

## The total of False positives are: 4568

cat("The total of False negatives are:", false_negatives2, "\n")

## The total of False negatives are: 688

cat("The total of True negatives are:", true_negatives2, "\n")

## The total of True negatives are: 85493

# Create a bar chart with the 4 types of results

results <- c(
  "True Positives" = true_positives2,
  "False Positives" = false_positives2,
  "False Negatives" = false_negatives2,
  "True Negatives" = true_negatives2
)

# Bar chart to show the AI's hits and errors
barplot(results,
        col = c("green", "red", "orange", "blue"),
        main = "AI Diagnosis Results 2",
        ylab = "Number of patients",
        las = 2)
```



It is reliable at first glance, but based on the numbers, there are high chances to NOT have the disease and still put you into the positive group.

4. Why are there so many false positives despite good specificity? Because of the chances to get it right are somewhat lower than a trustworthy code might happen (results might vary depending of luck)

5. Write a personal conclusion (minimum 5 lines) about what you learned regarding conditional probability and its application in AI. Conditional probability describes exactly the way an event occurs depending on the way something happens. And as a result, we need to know the chances something is going to happen. With this, we can take decisions based on the probability something happens; it also occurs in some way with AI decision taking, as the way chances might happen we can get a better way to confront something that someone might or might not get sick (In this case of course)

6. Export the RMarkdown file as a PDF.